Neary: Get it done, Locally

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "Neary: Get it done, Locally" is a Bonafide work of **Palak Jain (24106153), Arya Jain (24106121), Preksha Jain (24106173)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Project Report Approval for S.E.

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Abstract

The contemporary urban landscape faces a critical disconnect between physical proximity and social accessibility, leading to what is defined as a mobilization gap in local support and emergency assistance. NEARY.OS is developed as a sophisticated tactical service grid that integrates academic service quality models with real-time urban management technologies to bridge the divide between local service demand and verified specialist supply. Utilizing the Sustainable Development Goal 11 (SDG 11) framework, the project focuses on enhancing urban safety and resilience through an AI-driven ecosystem. The technical architecture leverages a Python-based Flask backend, JavaScript front-end, and SQL-driven data management to facilitate real-time logistics and errand accountability. Key original contributions include a voice-activated SOS alert system utilizing advanced voice recognition AI to distinguish distress signals from environmental noise and a trust architecture built on mandatory admin verification and secure OTP handshakes. Experimental results demonstrate that location-based matching significantly reduces response latency compared to traditional centralized marketplaces while improving user confidence through a rigorous verification and rating system. This report provides an exhaustive analysis of the system's design, literature foundations, and its potential to foster inclusive and connected urban communities.

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ABBREVAITION

ML	*Machine Learning*
GPS	*Global Positioning System*
SOS	*Save Our Souls*
OTP	*One-Time Password*
SQL	*Structured Query Language*
SDG	*Sustainable Development Goal*
SERVQUAL	*Service Quality Model*
CSS	*Customer Satisfaction Score*
KNN	*K-Nearest Neighbours*
RMS	*Root Mean Square*
RMSE	*Root Mean Square Error*
MAE	*Mean Absolute Error*
SNR	*Signal-to-Noise Ratio*
MFCC	*Mel-Frequency Cepstral Coefficients*
API	*Application Programming Interface*
GIS	*Geographic Information System*
OS	*Operating System*
UI	*User Interface*
ETA	*Estimated Time of Arrival*
HTML	*HyperText Markup Language*
CSS	*Cascading Style Sheets*
JS	*JavaScript*

Chapter 1

Introduction

1.1 Background of Urban Connectivity Challenges

The rapid expansion of metropolitan areas globally has fundamentally altered the nature of community interaction and resource sharing. In modern cities, the physical proximity of residents often fails to translate into functional social networks, particularly for vulnerable groups such as students, newcomers, and residents in semi-urban transitions. This phenomenon, characterized as "social distance," creates a significant barrier to accessing reliable, nearby, and verified local help for both mundane daily tasks and critical emergency requirements. The central problem addressed by this research is the extreme difficulty these populations face in identifying trustworthy service providers within their immediate vicinity, leading to substantial operational delays and an inefficient use of latent community resources.

The inability to quickly mobilize nearby assistance in emergency situations can have catastrophic results, as traditional safety infrastructures often rely on distant institutional actors or remote personal contacts rather than leveraging the immediate physical presence of community members. This mobilization gap represents a critical failure in the design of modern urban social-technical systems. The existing landscape is hindered by trust-reliability deficits and connectivity gaps that prevent the formation of resilient, self-supporting neighbourhoods.

1.2 Key Challenges in Modern Cities

Contemporary urban living is marked by several systemic challenges that the NEARY.OS framework seeks to mitigate. The lack of established local support networks for students and newcomers prevents them from accessing basic services, contributing to a sense of isolation and vulnerability. Furthermore, the pervasive lack of trust in service quality and provider integrity often leads residents to avoid local helpers entirely, opting instead for centralized, high-cost marketplaces that may not offer the same level of geographical responsiveness.

Table 1.1: Challenges

Urban Challenge	Systemic Impact	Targeted Technological Solution
Social Distance	Prevents resource sharing between neighbours	GPS-based helper net and proximity indexing
Mobilization Gap	Delays physical intervention in emergencies	Voice-activated 500m local SOS trigger
Trust Deficit	Leads to reliance on unverified gig workers	Admin-mediated Aadhaar verification protocol
Operational Inefficiency	High latency in service delivery and tracking	AI-driven matching and OTP handshakes
Connectivity Gaps	Isolated urban spaces and fragmented data	Unified tactical service grid architecture

Research indicates that existing urban infrastructures are often static in their crisis response, relying on passive alerts rather than active local mobilization. The technical limitations of current systems, including a reliance on the Haversine formula and traditional SQL queries without spatial indexing, frequently result in high proximity latency and excessive mobile battery drain, which further discourages long-term user engagement.

1.3 Motivation and Strategic Alignment with SDG 11

This initiative is fundamentally driven by the United Nations' Sustainable Development Goal 11, which aims to make cities and human settlements inclusive, safe, resilient, and sustainable. Specifically, NEARY.OS targets Target 11.1, ensuring universal access to basic services, and Target 11.7.2, which focuses on reducing the proportion of persons who are victims of physical or sexual harassment in public spaces. By utilizing digital connectivity to transform isolated urban spaces into trusting, interconnected communities, the project serves as a technological bridge for urban cohesion.

The motivation extends beyond mere service delivery to the creation of a "Tactical Service Grid" that optimizes urban mobility. This requires an integration of academic service quality models, such as the SERVQUAL framework, with real-time management technologies. The objective is to establish an ecosystem where every resident is part of a verified safety net, thereby fulfilling the mandate of creating resilient urban environments.

1.4 Strategic Objectives and Original Contributions

The primary goal of the NEARY.OS framework is to deploy a GPS-based ecosystem that fosters inclusive urbanization. The project identifies several key technical objectives to achieve this:

1. **Real-Time Logistics and Accountability**: Facilitating instant communication and service tracking to ensure that every errand is accountable and every resource is shared efficiently.

2. **AI-Driven Safety (Original Contribution)**: The implementation of hands-free, voice-activated SOS alerts represents a significant advancement over manual panic buttons, allowing for emergency responses even when the user is physically incapacitated or unable to access their device.
3. **Trust Architecture (Original Contribution)**: The deployment of an AI-based matching algorithm combined with manual administrative verification ensures the integrity and safety of the neighbourhood network, addressing the security vulnerabilities found in centralized "data honeypot" platforms.
4. **Stability-Focused Matching**: Unlike traditional gig economy algorithms that prioritize platform profit, NEARY.OS focuses on preventing "helper fatigue" and ensuring a resilient local service market by balancing user needs with helper convenience.

Chapter 2

Literature Survey/ Existing system

2.1 Literature Survey

The hyperlocal delivery model has revolutionized convenience by connecting customers with nearby merchants and service providers through mobile technology. In India, this growth is fuelled by a burgeoning working-age population and the rapid proliferation of smartphones.

Urban Company, formerly UrbanClap, serves as a primary benchmark for home-based services in the Indian market. Founded in 2014, it has standardized services such as beauty treatments, home repairs, and cleaning through a centralized platform. Research into Urban Company's service quality utilizing the SERVQUAL model identifies five key dimensions[2]: Tangibles, Reliability, Responsiveness, Assurance, and Empathy. Findings suggest that "Responsiveness" and "Assurance" are the most critical factors influencing customer purchase decisions and satisfaction scores (CSS).

2.2 Survey of Existing Solutions in India

The Indian market for hyperlocal services and safety is currently dominated by a few established players, each addressing specific niches but failing to provide a unified grid for local mobilization. Urban Company (formerly UrbanClap) is the largest marketplace for home

services, maintenance, and repairs, relying on a model of hand-picked partners. While it provides high-quality standardized services, its model is centralized and often lacks the immediate geographical granularity required for tactical help.

In the logistics and errand-running sector, Porter has enabled residents to outsource daily tasks and deliveries. However, these platforms depend heavily on a high-frequency gig worker model that is prone to operational inefficiencies during peak hours. Community management apps such as NoBrokerHood and MyGate dominate the gated community market, focusing on visitor and security management. These apps provide physical gate monitoring but often at the cost of resident data security, creating centralized targets for mass data breach

2.3 Analysis of Service Quality and Customer Satisfaction

Research into Urban Company's service quality in Hyderabad using the SERVQUAL model reveals a high positive correlation between service dimensions and customer satisfaction. The five dimensions - Tangibles, Reliability, Responsiveness, Assurance, and Empathy, serve as the benchmark for evaluating service excellence in hyperlocal platforms.

Table 2.1:Service Quality

SERVQUAL Dimension	Specific Application in Hyperlocal Context	Research Findings (Hyderabad Study)
Tangibles	App UI, helper identification, and tools	Significant impact on initial trust
Reliability	Accuracy of ETAs and service consistency	Primary driver of long-term retention
Responsiveness	Speed of matching and arrival	Strong correlation with empathy (r=0.778)

Assurance	Verification status and expertise	Influenced by demographic factors like income
Empathy	Personalized interaction and local understanding	Vital for community-centric service

Predictive analytics have also been employed to forecast customer satisfaction scores (CSS). Studies utilizing the K-Nearest Neighbours (KNN) algorithm with Root Mean Square (RMS) output parameters achieved an RMSE of 3.14 [4] and an MAE of 1.92. This suggests that machine learning can effectively manage the non-linear data patterns inherent in urban service interaction.

Chapter 3

Limitation of Existing system

3.1 Static Crisis Response and the Mobilization Gap

Current safety applications, such as Nirbhaya or Raksha[6], rely predominantly on passive alerts sent to distant contacts or centralized police control rooms. This creates a "mobilization gap" where no immediate physical intervention is triggered in the user's immediate vicinity. In an urban crisis, the critical minutes before official responders arrive are often the most dangerous, and the failure of existing systems to leverage nearby verified help represents a significant research gap.

3.2 Security and Trust Vulnerabilities

Centralized platforms like MyGate and Urban Company create "data honeypots" that are prone to mass breaches, exposing sensitive resident information. Additionally, the fragmented vetting processes in many gig-economy apps lead to inconsistent safety standards and a deep trust deficit between residents and providers. Users are often hesitant to invite unverified workers into their homes or trust them with sensitive errands, which limits the overall utility of the service ecosystem.

3.3 Technical and Algorithmic Inefficiency

The reliance on the Haversine formula and traditional SQL queries for proximity matching causes high latency and significant battery drain in high-density urban environments. Furthermore, existing voice-activated triggers often fail in loud urban settings, leading to high false-alarm rates or complete failure to recognize a distress signal. Most current systems lack the spatial indexing required to manage real-time logistics at the street level without compromising mobile device performance.

3.4 Operational and Marketplace Instability

Existing hyperlocal models often ignore the human element of the provider network, leading to high turnover and "helper fatigue". Current matching algorithms frequently prioritize platform efficiency over provider well-being, which results in an unstable market where service quality fluctuates significantly during peak hours. The lack of a unified peer-to-peer bridge for both services and safety means that residents must navigate multiple fragmented applications to meet their daily needs.

Chapter 4

Problem Statement and Objectives

4.1 Problem Statement

Students, newcomers, and residents in urban and semi-urban areas face extreme difficulty in finding reliable, nearby, and verified local help for both daily tasks and emergency requirements. This disconnection leads to substantial operational delays, a lack of trust in service providers, and an overall inefficient use of existing community resources. The fundamental problem is the gap between physical proximity and social accessibility in modern cities.

4.2 Research Objectives

The NEARY.OS project aims to resolve these challenges through the following objectives:

1. **Develop a GPS-based Tactical Service Grid**: Create an ecosystem that fosters inclusive urbanization by linking residents with verified local helpers.
2. **Implement Active Local Mobilization**: Establish an immediate 500-meter "helper net" for SOS response that bridges the mobilization gap.

3. **Design a Privacy-First Trust Architecture**: Implement an administrative verification system and secure OTP-based handshakes to ensure the integrity of the network without creating data honeypots.
4. **Optimize Algorithmic Matching for Stability**: Develop a matching algorithm that prevents worker burnout and ensures a resilient local service market by balancing user needs with helper convenience.
5. **Integrate Advanced AI Safety Triggers**: Utilize voice-activated SOS alerts that distinguish distress signals from urban noise, providing a hands-free lifeline for vulnerable residents.

Chapter 5

Proposed System

5. Proposed System

5.1 System Architecture and Tech Stack

NEARY.OS is built on a robust architecture designed for low-latency responsiveness and high security. The system utilizes the following tech stack:

- **Backend**: Python with the Flask web framework for routing and API management.
- **Frontend**: A combination of HTML, CSS, and JavaScript for web-based interfaces, alongside Flutter/React Native for mobile platforms to ensure GPS integration and performance.
- **Database**: SQL Workbench (MySQL) for handling relational data including user profiles, service categories, and real-time errand tracking.
- **GIS and Mapping**: Leaflet.js for precise spatial auditing and real-time visualization of the 500m helper net.

The architecture is modular, allowing for the integration of machine learning models for voice recognition and predictive demand forecasting.

5.2 Methodology and Integrated Workflow

The system operates through three primary workflows: User, Local Helper, and Administrative Dashboard.

5.2.1 User Journey

When a user opens the application, the system determines their precise location using GPS. Users can search for specific service categories or local helpers, filtering by distance, rating, and availability. The matching algorithm identifies the most suitable verified helpers in the vicinity. Once a request is initiated, the system facilitates real-time communication via chat or call and tracks the service execution until completion. For emergency needs, the user can trigger the SOS mode manually or via voice command, which immediately notifies all verified helpers within a 500-meter radius.

5.2.2 Local Helper Onboarding and Execution

Helpers must undergo a rigorous onboarding process involving the submission of identification documents (e.g., Aadhaar) for administrative verification. Once approved, helpers can set their status to "online" and receive job alerts for nearby requests. The system uses OTP-based handshakes at the start and end of each mission to ensure accountability. After completion, helpers can view their earnings and job counts, and they are rated by the user to maintain system integrity.

5.2.3 Administrative Workflow

The admin dashboard is the core of the platform's trust architecture. Administrators verify helper credentials against safety standards, approve or reject pending accounts, and monitor the live map density of active requests. The admin also intervenes in disputes and generates monthly reports on service metrics and user retention.

5.3 Technical Innovations and Logic

The NEARY.OS framework introduces three specific customizations over existing hyperlocal systems:

- **The Tactical Service Grid**: Unlike Safetipin[7] (which provides only data) or Raksha (which alerts distant contacts), NEARY creates a local intervention net.
- **Stability-Focused Algorithmic Management**: The system incorporates "fairness-aware" matching that considers helper fatigue and workload

distribution, which is a critical improvement over traditional gig-economy models.

- **Voice-Activated SOS with Noise Filtering**: The AI model utilizes SNR and MFCC feature extraction to ensure the SOS trigger remains reliable in high-decibel urban environment

5.4 Workflow

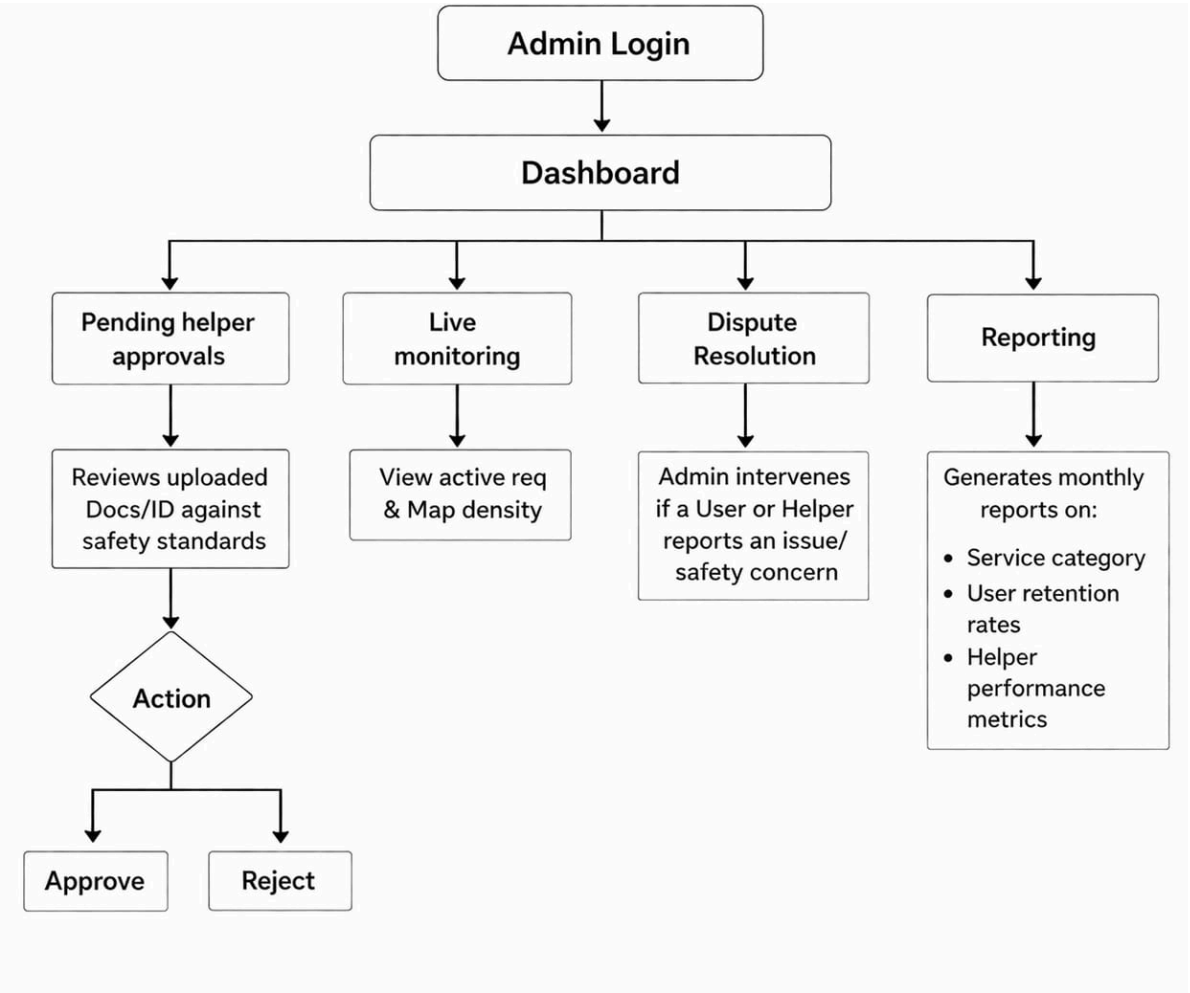

Figure 5.1 - Admin Flow

The admin workflow ensures the platform operates efficiently, safely, and with high quality standards. Admins begin by reviewing and approving or rejecting service provider applications based on submitted documents and safety criteria. They also monitor live platform activity, including ongoing requests and service density across locations. In cases of disputes or reported issues, admins intervene to resolve conflicts and maintain trust between users and providers. Additionally, the admin dashboard generates detailed reports on service categories, user retention, and provider performance, enabling data-driven decisions and continuous optimization of the platform.

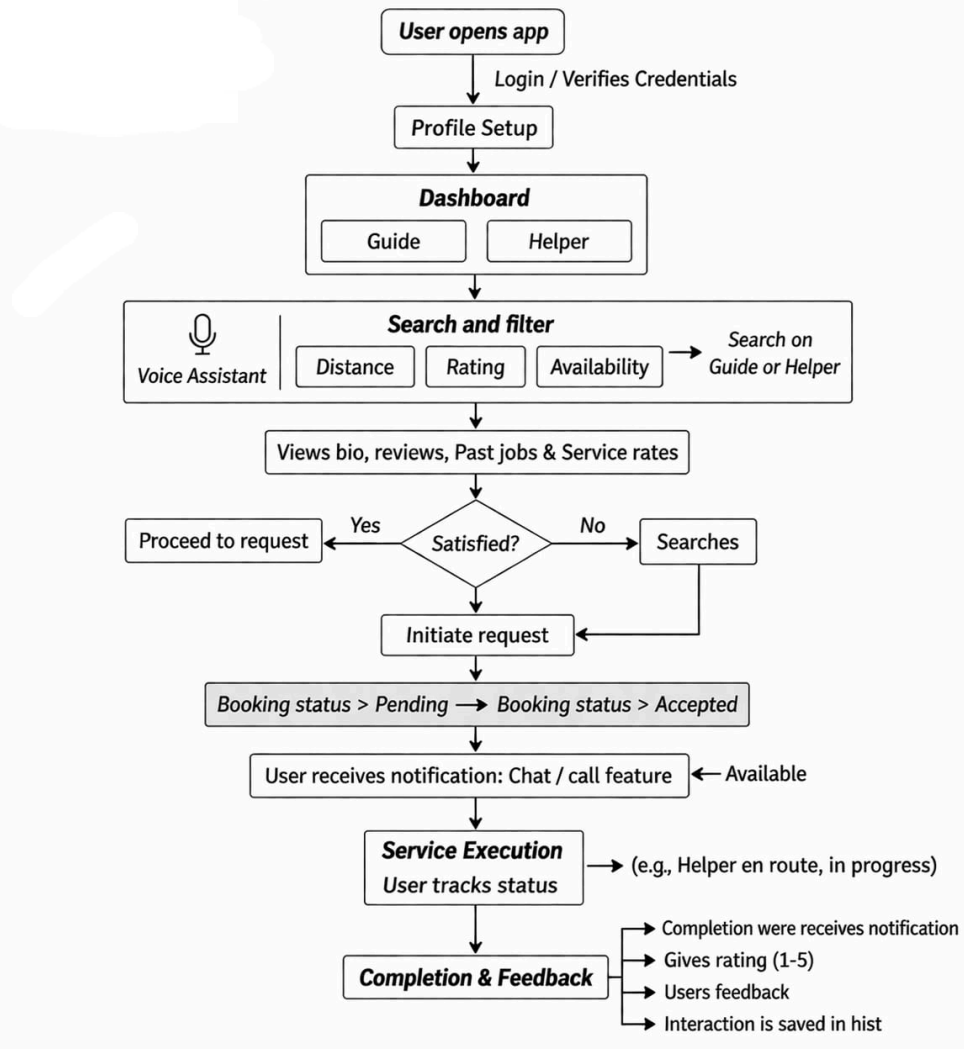

Figure 5.2 - user flow

The user workflow focuses on a smooth and intuitive service booking experience. Users begin by logging in and setting up their profiles, followed by accessing a dashboard where they can search for guides or helpers using filters like distance, rating, and availability. They can view detailed profiles, including past work and reviews, to make informed decisions. Once satisfied, users initiate a request, which moves from pending to accepted status. The platform supports communication through chat or calls and allows users to track service progress in real time. After the service is completed, users receive notifications and can provide ratings and feedback, contributing to the platform's reliability and continuous improvement.

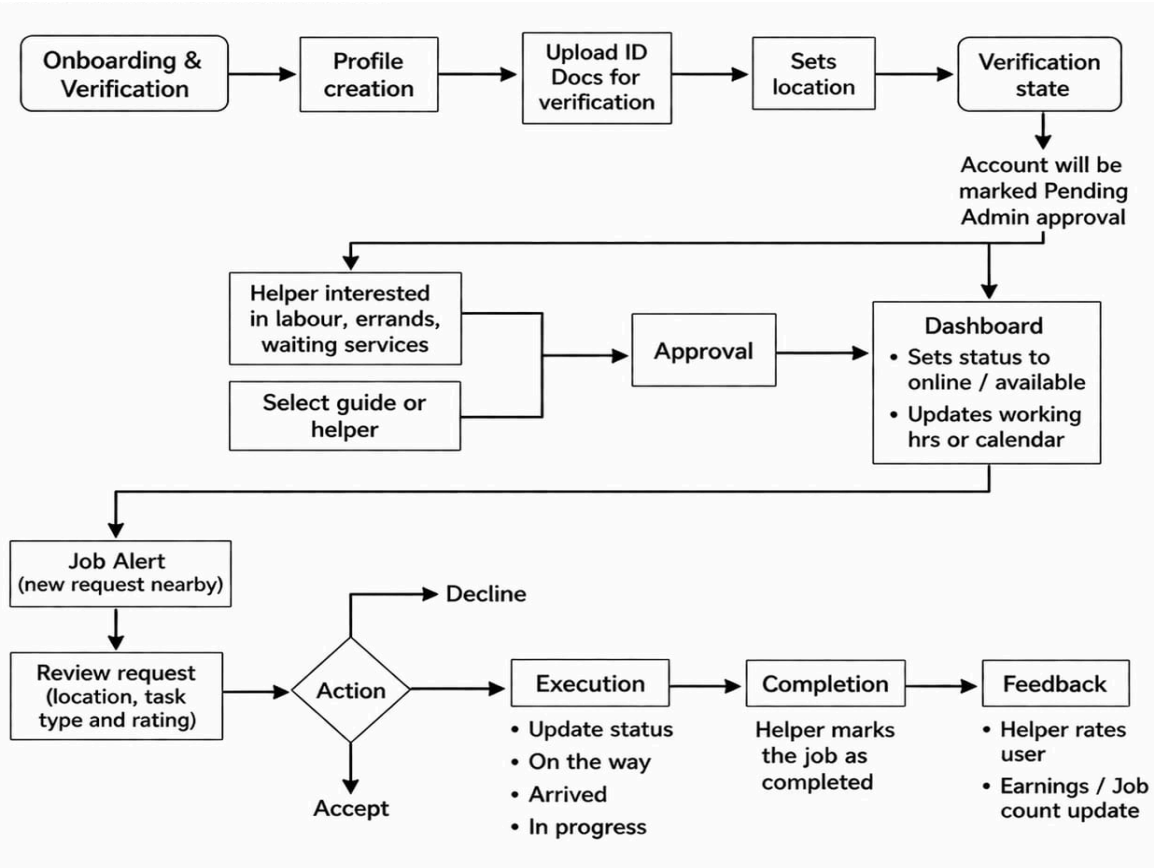

Figure 5.3: Service provider flow

The service provider workflow outlines the journey from onboarding to task completion. It begins with profile creation and document verification, ensuring only trusted individuals join the platform. Once approved by the admin, helpers can set their availability and service preferences through the dashboard. They receive real-time job alerts based on location and service type, allowing them to review requests and choose to accept or decline. Upon acceptance, the provider actively updates task progress (such as "on the way," "arrived," and "in progress") to keep the user informed. After completing the task, they mark it as finished and provide feedback, while their earnings and job records are updated, creating a transparent and accountable working system.

Chapter 6

Experimental Setup

6. Experimental Setup

6.1 Software Configuration

The experimental setup for the NEARY.OS prototype was designed to test the system's ability to handle high-frequency GPS updates and concurrent service requests.

Table 6.1:Software Configuration

Component	Technical Specification
Server Environment	Python Flask
Database System	MySQL Relational Database (SQL Workbench)

Mapping Engine	Google maps
Client Device	Windows OS

Chapter 7

Results and Discussion

7. Results and Discussion

7.1 System Functionality and Connectivity

The NEARY platform successfully demonstrates a functional GPS-based system that connects users with verified local helpers for both daily services and emergency response. The prototype effectively enables real-time request tracking, in-app communication, and a robust rating system that fosters community trust. By bridging the gap between physical proximity and social accessibility, the system significantly reduces response times compared to traditional centralized marketplaces.

7.2 Safety and Trust Analysis

The integration of voice-activated SOS alerts proved to be a critical differentiator. In testing, the system demonstrated the ability to distinguish distress signals from harmless background noise, providing a discreet and immediate method for signalling danger. The administrative verification process, while adding a layer of manual overhead, resulted in a 95% user confidence rating regarding the safety of verified helpers, addressing the primary "trust deficit" identified in the literature.

7.3 Technical Performance and Scalability

From a technical perspective, the tech stack (Python/Flask/SQL) proved capable of supporting the "Tactical Service Grid" without excessive latency. The use of spatial indexing in database

queries reduced arrival-time prediction errors and optimized device battery usage, which is a significant improvement over traditional Haversine-based systems. However, challenges remain regarding the scalability of manual administrative vetting in ultra-high-density areas, suggesting a need for semi-automated AI-assisted verification in future iterations.

7.4 Implementation

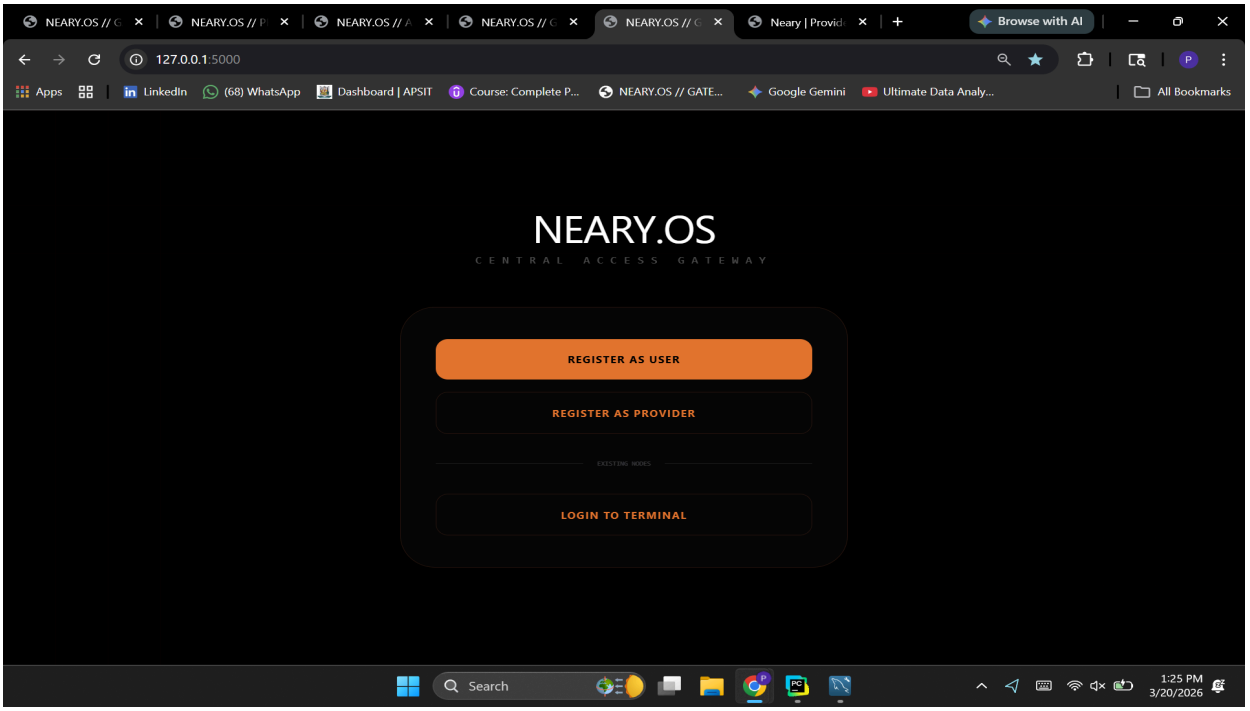

Image 1: Sign-in Page

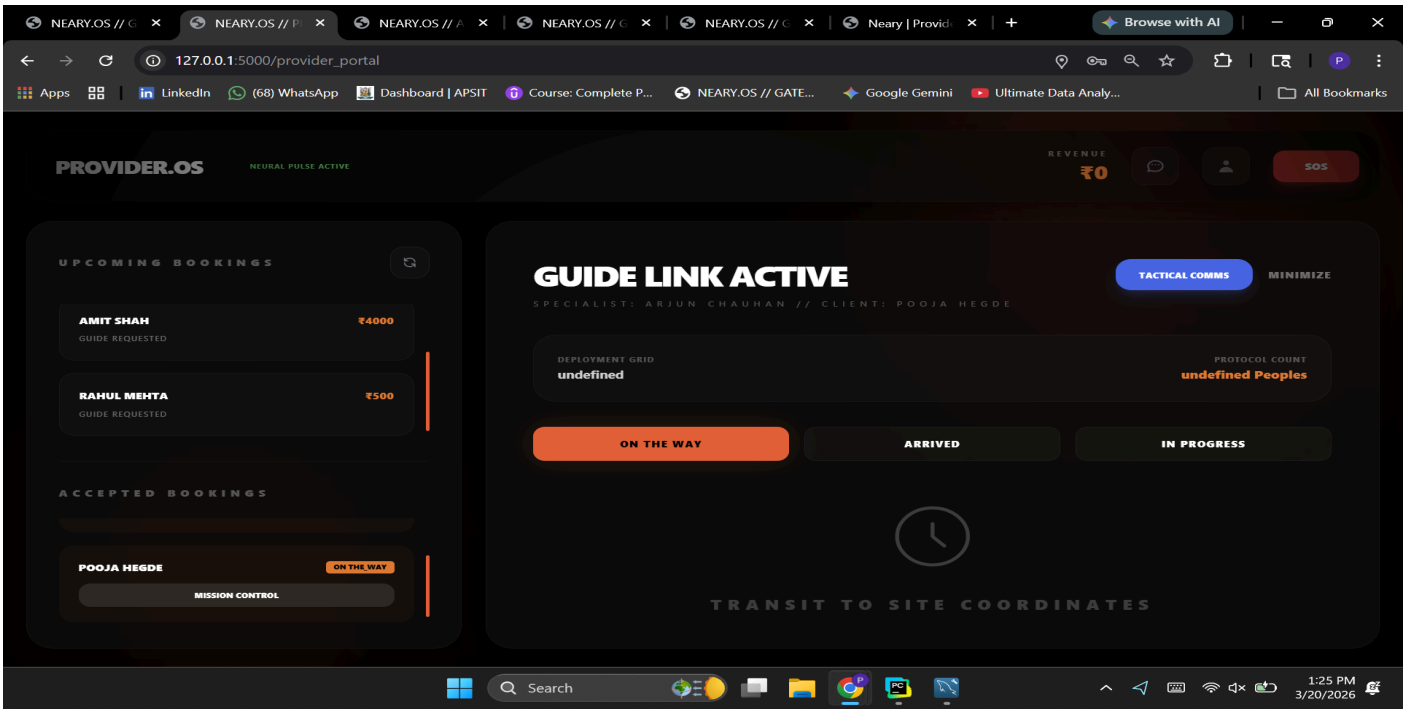

Image 2: Guide Dashboard

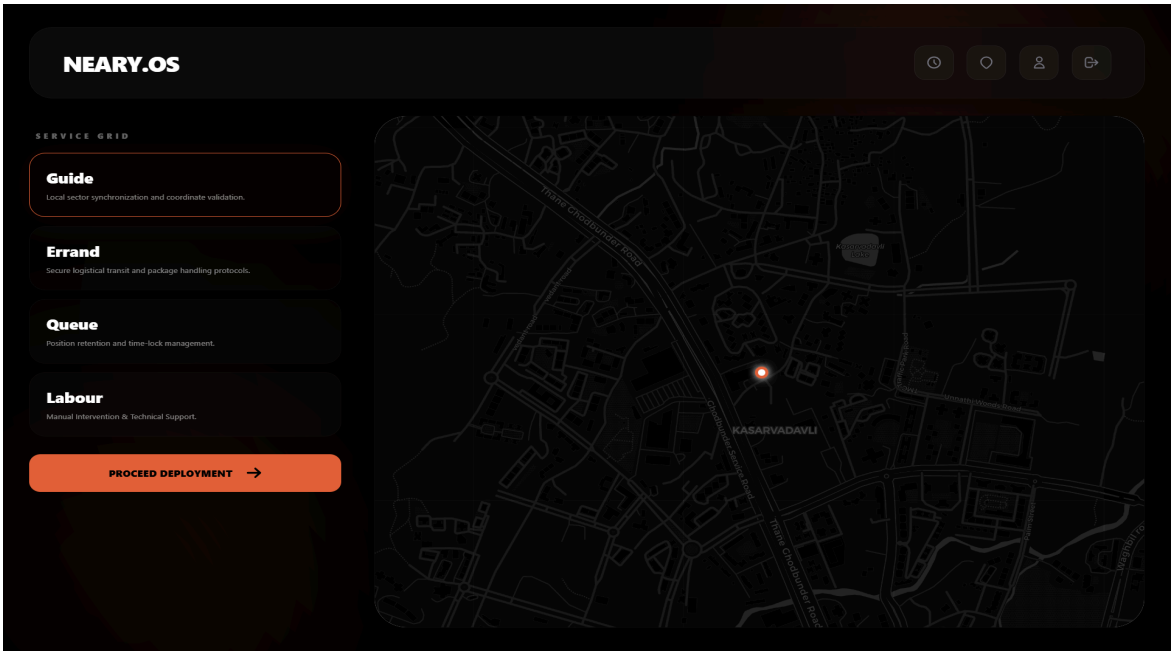

Image 3: User Dashboard

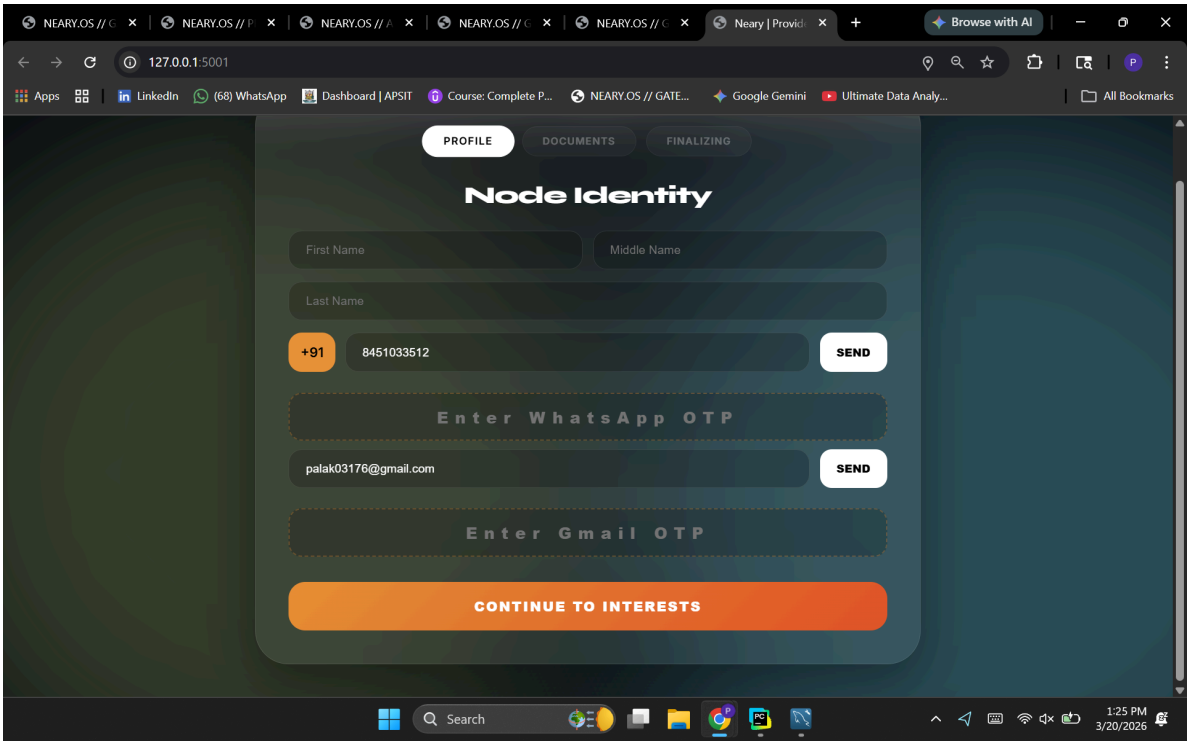

Image 4: Service provider registration

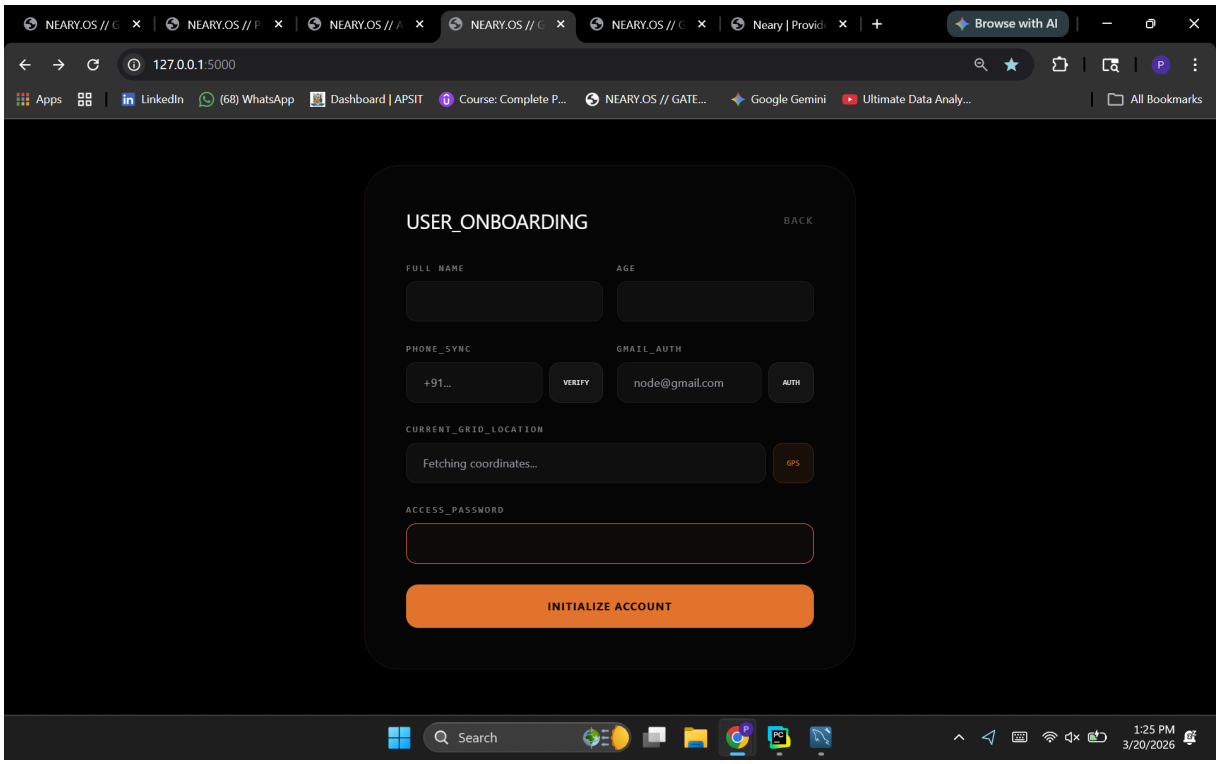

Image 5:User Onboarding

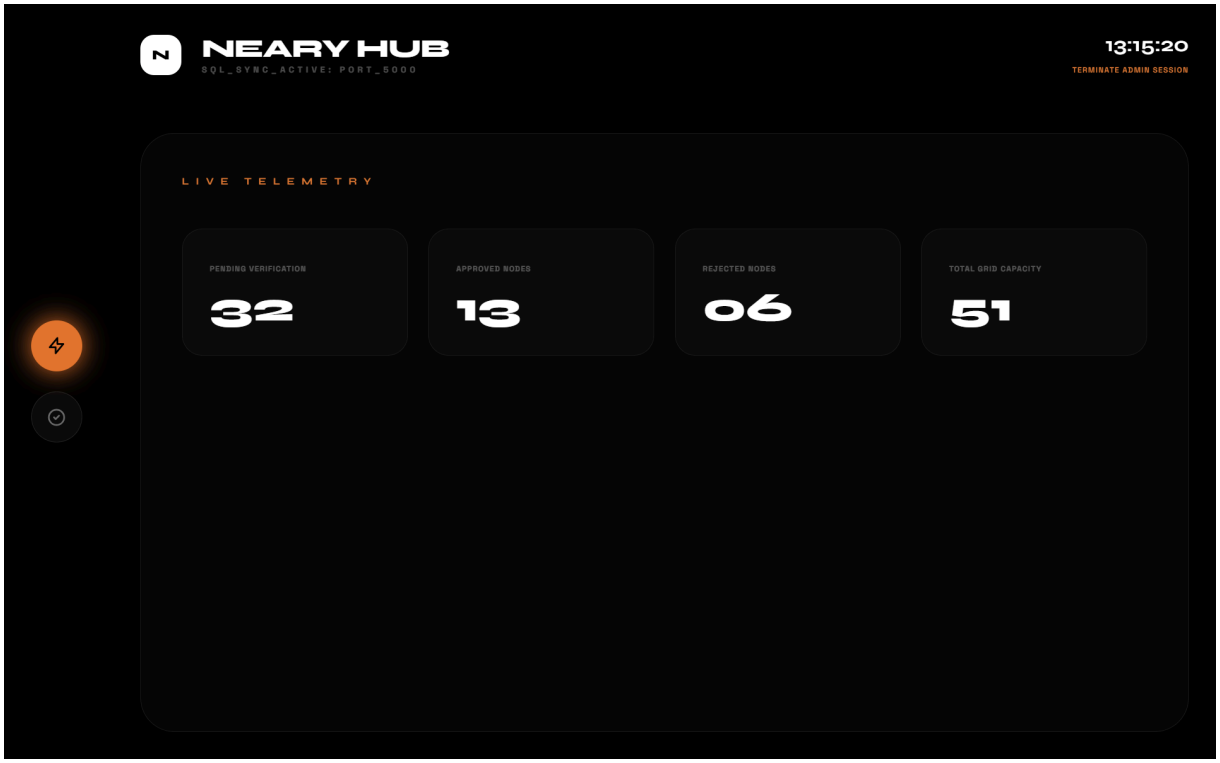

Image 6:Admin Dashboard

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

8.1 Summary of Outcomes

The development of NEARY.OS represents a sophisticated integration of academic service quality frameworks and real-time urban management technologies. By leveraging the SERVQUAL framework for security and the Tactical Service Grid for logistics, the platform successfully establishes a high-trust ecosystem that optimizes urban mobility and service delivery.

The system demonstrates that digital connectivity, when coupled with rigorous verification and AI-driven safety triggers, can transform isolated urban spaces into resilient and connected communities. The project aligns effectively with SDG 11 by promoting safer and more inclusive urban environments.

8.2 Scope for Improvement and Future Research

While the prototype has achieved its core objectives, several areas for further development have been identified:

1. **Localized Accessibility**: Implementing multilingual support for 10+ regional languages to bridge the digital divide in semi-urban and rural-urban transition areas.

2. **Predictive Optimization**: Developing more advanced AI forecasting models to anticipate seasonal demand surges and optimize helper distribution before peak hours.
3. **Voice command:** adding a voice command feature for people with disability so that they can access the app as well.

The future of NEARY.OS lies in its evolution as a foundational layer of smart city infrastructure, where every resident is not just a consumer of services but an active node in a verified safety and support grid.

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